

MODULE 10 · DOMAIN 3 - CLOUD TECHNOLOGY AND SERVICES

Global Infrastructure

Regions, Availability Zones, edge locations, and choosing between them.

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There is a **hierarchy**. Most questions are asking which level of it you should be operating at.

The levels of the hierarchy

LEVEL	WHAT IT IS
Region	A geographic area containing multiple Availability Zones
Availability Zone	One or more discrete data centres, isolated from the others
Local Zone	An extension of a Region placed nearer a population centre
Edge location	A caching site for CloudFront, far more numerous than Regions
AWS Outposts	AWS-managed racks installed in your own data centre

ON THE EXAM

Regions and AZs are where you run things. Edge locations are where content is cached. Outposts is AWS hardware on your premises.

Region or Availability Zone?

AWS REGION

A geographic area

- Named for a place, such as eu-west-2
- Contains multiple Availability Zones
- You choose one when you create most resources
- Data does not leave it unless you move it

AVAILABILITY ZONE

Isolated infrastructure inside a Region

- Discrete data centres with their own power and cooling
- Physically separated but close together
- Linked by high-bandwidth, low-latency private networking

THE TIEBREAKER

Geography and data residency means Region. Surviving a failure and staying available means Availability Zone.

What an Availability Zone actually is

- ◆ **One or more data centres** with independent power, cooling and physical security.
- ◆ Separated by enough distance that one disaster is unlikely to affect two.
- ◆ Close enough that replication between them stays fast enough to be synchronous.
- ◆ **Connected by private, high-bandwidth, low-latency links**, not the public internet.
- ◆ Deploying two instances in the same AZ gives you redundancy but not high availability.

ON THE EXAM

The design trade-off is deliberate: far enough apart to fail independently, close enough to replicate synchronously.

Availability Zone or edge location?

AVAILABILITY ZONE

Where your workload runs

- Compute, storage and databases live here
- Used to survive infrastructure failure
- A handful per Region
- You choose which ones to deploy into

EDGE LOCATION

Where content is cached

- CloudFront caches copies close to users
- Used to reduce latency, not to survive failure
- Far more numerous, in many more cities
- You do not deploy workloads here

THE TIEBREAKER

High availability means Availability Zones. Delivering content quickly to distant users means edge locations.

Edge locations in practice

- ◆ **Amazon CloudFront caches content** at edge locations so users fetch it from nearby rather than from the origin.
- ◆ Regional edge caches sit between edge locations and the origin, holding content that is too big or too infrequent for the edge.
- ◆ Amazon Route 53 and AWS Global Accelerator also use the edge network.
- ◆ **AWS Shield Standard and AWS WAF operate at the edge** when used with CloudFront.
- ◆ Caching at the edge does not make your origin more available. If the origin is down, uncached content fails.

ON THE EXAM

Edge locations reduce latency and absorb load. They are not a high availability strategy for the origin.

Local Zones, Outposts and Wavelength

- ◆ **AWS Local Zones** place compute and storage nearer a large population centre for single-digit millisecond latency.
- ◆ **AWS Outposts** installs AWS-managed racks in your own data centre, running AWS services locally.
- ◆ Outposts is the answer when data must physically remain on your premises, or latency to the Region is unacceptable.
- ◆ Both connect back to a parent Region and are managed with the same tools.
- ◆ AWS Wavelength serves 5G networks, and it is not on the CLF-C02 in-scope list.

ON THE EXAM

Must stay in our building means Outposts. Very low latency in a specific city means Local Zones.

How to choose a Region

01

Compliance

Data governance and legal requirements. This is non-negotiable and comes first.

02

Latency

Proximity to your users.
Distance costs milliseconds.

03

Service availability

Not every service exists in every Region.

04

Pricing

The same resource costs different amounts in different Regions.

ON THE EXAM

If a scenario mentions a legal or regulatory requirement about where data may live, compliance overrides all three others.

Multi-AZ or multi-Region?

MULTIPLE AVAILABILITY ZONES

Within one Region

- Survives the loss of a data centre
- The standard answer for high availability
- Low latency between zones
- Usually little or no extra complexity

MULTIPLE REGIONS

Across geographies

- Survives the loss of an entire Region
- Used for disaster recovery and global latency
- Required for some data residency rules
- More cost, more complexity, more data transfer

THE TIEBREAKER

High availability means multi-AZ. Disaster recovery, global users, or data sovereignty means multi-Region.

Global and Regional services

- ◆ **Global:** IAM, Amazon Route 53, Amazon CloudFront, AWS Organizations.
- ◆ **Regional:** Amazon EC2, Amazon VPC, Amazon RDS, Amazon DynamoDB, AWS Lambda.
- ◆ Regional resources must be created again in each Region you want them in.
- ◆ Amazon S3 bucket names are globally unique, but the bucket itself lives in one Region.
- ◆ An AMI is Regional. Copying it to another Region is a deliberate step.

ON THE EXAM

Launching in a second Region means re-creating Regional resources. Global services need no duplication.

Reverse lookup

IF THE SCENARIO SAYS	THE ANSWER IS
Stay available if a data centre fails	Multiple Availability Zones
Survive the loss of an entire Region	Multiple Regions
Data must remain within a country's borders	Choose the Region; compliance decides
Users worldwide complain of slow page loads	CloudFront and edge locations
Data must stay in our own building	AWS Outposts
Single-digit millisecond latency in one city	AWS Local Zones

ON THE EXAM

Six phrasings, six answers. Almost every global infrastructure question is one of these in different words.

A company runs its web application on two EC2 instances behind a load balancer. Both instances are in the same Availability Zone. During a power event affecting that zone, the application became completely unavailable.

The company wants the application to stay available if a single Availability Zone fails, at the lowest reasonable cost and complexity.

What should the company do?

- ☐ A Add a third instance in the same Availability Zone
- ☐ B Distribute instances across multiple Availability Zones in the Region
- ☐ C Deploy a copy of the application in a second Region
- ☐ D Put Amazon CloudFront in front of the load balancer

◆ PAUSE AND DECIDE

What should the company do?

- A

 Add a third instance in the same Availability Zone
More redundancy against instance failure, none against zone failure. The same event takes all three down.
- B

 Distribute instances across multiple Availability Zones in the Region
Zones fail independently. This is the standard high availability answer, with minimal added complexity.
- C

 Deploy a copy of the application in a second Region
Would work, but solves Region-level failure at far greater cost and complexity than the requirement asks for.
- D

 Put Amazon CloudFront in front of the load balancer
Reduces latency and caches content. If the origin's only zone is down, uncached requests still fail.

THE SIGNAL

Single Availability Zone failure plus lowest reasonable cost. Multi-AZ is the exact fit; multi-Region overshoots.

A German insurer must keep all customer data physically within Germany to satisfy its regulator. It also wants low latency for its staff, who are all in Frankfurt and Munich, and would prefer the cheapest option available.

One Region satisfies the residency rule. A different Region is cheaper.

Which factor should decide the Region?

- ☐ A Pricing, because the company prefers the cheapest option
- ☐ B Latency, because staff are concentrated in two cities
- ☐ C Compliance, because the regulator requires data to stay in Germany
- ☐ D Service availability, because not all services exist everywhere

◆ PAUSE AND DECIDE

Which factor should decide the Region?

- A

 Pricing, because the company prefers the cheapest option
Prefer is a preference. A regulator's requirement is not something a preference can outrank.
- B

 Latency, because staff are concentrated in two cities
A genuine benefit, but it does not conflict here and it cannot override a legal constraint.
- C

 Compliance, because the regulator requires data to stay in Germany
A legal requirement about where data may reside eliminates every Region that fails it, before anything else is weighed.
- D

 Service availability, because not all services exist everywhere
Real, but the scenario never suggests a needed service is missing.

THE SIGNAL

Must and would prefer are doing the work. One is a constraint, the other is a wish.

A media company hosts video and images in a single Region in the United States. Users in Australia and Japan report that pages take several seconds to load, although the application itself is performing normally and is never unavailable.

The company wants faster page loads for those users without moving or duplicating the application.

What should the company use?

- ☐ A Deploy the application into Regions in Australia and Japan
- ☐ B Add more Availability Zones in the United States Region
- ☐ C Amazon CloudFront
- ☐ D AWS Outposts in Australia and Japan

◆ PAUSE AND DECIDE

What should the company use?

- A

 Deploy the application into Regions in Australia and Japan
Duplicating the application is exactly what the scenario rules out in its final sentence.
- B

 Add more Availability Zones in the United States Region
Improves availability, which is not the problem. Nothing has been unavailable.
- C

 Amazon CloudFront
Caches the content at edge locations near those users. Latency falls and the application is untouched.
- D

 AWS Outposts in Australia and Japan
Puts AWS hardware in your own facilities. The company has no facilities there, and this is not a residency problem.

THE SIGNAL

Slow but never unavailable is a latency problem, not an availability one. Static content plus distant users means CloudFront.

A manufacturer runs machinery in a factory that must be controlled with very low latency, and a national regulation requires the process data to remain physically inside the factory site. The nearest AWS Region is several hundred kilometres away.

The company wants to use AWS services and tooling for this workload.

Which service meets these requirements?

- ☐ A AWS Local Zones
- ☐ B AWS Outposts
- ☐ C Amazon CloudFront
- ☐ D An additional Availability Zone

◆ PAUSE AND DECIDE

Which service meets these requirements?

A AWS Local Zones

The strongest distractor. It solves the latency half, but a Local Zone is an AWS facility, not your factory.

C Amazon CloudFront

Caches content for delivery to users. It does not run a control workload or hold data on premises.

B AWS Outposts

AWS-managed racks inside the factory. Data stays on site and latency is local, using the same AWS tooling.

D An additional Availability Zone

Still hundreds of kilometres away, and still an AWS facility. Neither requirement is met.

THE SIGNAL

Two requirements: low latency, and data physically on your own site. Only Outposts satisfies the second.

What to carry out of this module

- ◆ **Region is geography. Availability Zone is isolation. Edge location is caching.**
- ◆ **Multi-AZ is high availability. Multi-Region is disaster recovery, global reach or sovereignty.**
- ◆ Choosing a Region: compliance, then latency, then service availability, then price.
- ◆ Outposts means your building. Local Zones means a nearby city. Both keep AWS tooling.
- ◆ Edge caching does not make the origin more available. Latency and availability are different problems.

ON THE EXAM

Must and required are constraints. Prefer and want are wishes. Constraints eliminate options first.